

Q1. Could you please explain me the difference between online and offline UPS systems with the help of a diagram?

Iyer Parameswar Narayanan

Through e-mail

A1. All of us take mains power protection and backup for granted and use mains power supply very casually without even thinking about the damage it can cause to our sophisticated and sensitive electronic equipment. Ordinary household appliances such as tubelights, fans, TV and fridge are usually unaffected by AC mains supply fluctuations. But

er fast enough to prevent computers from rebooting. The switchover time is usually two to 20 milliseconds.

As mains power is bypassed to the load during normal power supply, these UPS systems are called 'offline.' They are also known as standby UPS systems as they are used only during power failures.

During normal power supply, offline UPS systems also charge the connected batteries while supplying power to the load. The bypassing of power results in exposing the load to electrical noises and surges. Therefore offline UPS systems are not recommended for

very sensitive and critical electronic devices. As online UPS systems always provide power from the battery, the load gets clean power regardless of any power problem. Therefore online UPS systems are recommended for sensitive and critical electronic devices.

Online UPS systems provide a layer of insulation or electric firewall. Any change in the input voltage or frequency does not affect the output voltage and frequency. Therefore the load is shielded from all power problems.

Online UPS systems also allow connection of large battery packs, allowing a good power backup in excess of four hours. Many online UPS models also offer input power factor correction.

Most offline

UPS systems available in the market provide a sine wave output during normal power supply. When they switch to their internal DC-AC inverter, they

Q2. What are TRIM analysers?

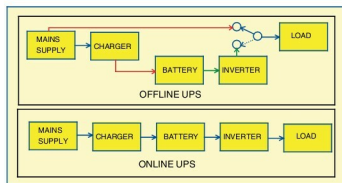
V. Harshita

Through e-mail

A2. TRIM stands for tissue resonance impedance monitor. This is an extensive technique that can be used to assess the body composition of humans.

As defined in a paper published by P.J. Chadwick and N.H. Saunders from University College of Swansea, TRIM analyser is an electromagnetic body composition analyser that holds considerable promise as a rapid, safe, reliable, non-invasive and inexpensive predictor of lean body mass.

It takes advantage of the fact that the configuration of eddy currents induced in the human body when subjected to an external electromagnetic field reflects the distribution of conducting and non-conducting tissues. Fat tissues are non-conducting, while fat-free tissues are conducting.



Comparison between online and offline UPS systems

for computers and other sophisticated electronics equipment, a clean and stable uninterrupted power supply is highly recommended.

Among all the household appliances, computers are probably the most sensitive to power disturbances and failures. Interruptions in power supply may cause corruption or loss of important data. Even the entire system can malfunction or fail to work. For such sensitive equipment, uninterrupted power supply (UPS) is a must.

There are mainly two types of UPS systems available in the market: offline and online.

Offline UPS systems. Offline UPS systems run appliances directly from mains power as long as it is available. However, during power failures and other power problems, they switch from AC mains power to battery power

may provide a square wave, modified square wave or quasi-sine wave. In many cases, your equipment may appear to operate normally on these waveforms but may get damaged by them over time. These UPS typically provide battery backup for 5 to 15 minutes when loaded to their full output capacity.

Online UPS systems. In case of online UPS systems, appliances are powered from batteries all the time. Batteries in the UPS system are continuously charged from mains power. The stored battery power is used to power the load.

The mains power is converted from AC to DC to charge the battery, and the battery power is converted from DC to AC to power the load. Therefore online UPS systems are also known as 'double-conversion' UPS systems.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/I, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender